

Number Cross

Jane Street, August 2014

1	2	3		4	5	6		7	8	9
10				11				12		
13				14			15			
		16	17							
18	19					20		21	22	23
24							25			
26				27		28				
			29		30					
31	32	33						34	35	36
37				38				39		
40				41				42		

Place the digits 1 through 9 (no zeroes) so that all clues are satisfied.
No digit is repeated within any one grid entry, and no grid entry is used more than once in the puzzle.

Answer: the sum of the eight 5-digit entries — 18a, 20a, 26a, 28a, 4d, 6d, 27d, 28d.

ACROSS

1. (3) A square number
4. (3) A square number
7. (3) A square number
10. (3) Smaller than 23-down
11. (3) A square number
12. (3) A square number
13. (3) The smallest number in this grid
14. (7) Its digits sum to 42
16. (6) Divisible by 6
18. (5) 37-across + 6-down
20. (5) Whether read forwards or backwards,
the largest 5-digit number in this
grid
24. (4) Has the same digits as 5-down
25. (4) 42-across + 7-down
26. (5) Its digits sum to 26
28. (5) Its digits are ascending
29. (6) 4-across times 35-down
31. (7) Divisible by 3
34. (3) Divisible by 3
37. (3) Its digits are ascending
38. (3) 39-across minus 13-across
39. (3) Twice a prime number
40. (3) Greater than 1-across
41. (3) Divisible by 25
42. (3) An odd number

DOWN

1. (3) A square number
2. (3) Divisible by 11
3. (7) The largest number in this grid
4. (5) Its digits sum to 16
5. (4) A multiple of 9-down
6. (5) Its digits sum to 35
7. (3) A square number
8. (3) A square number
9. (3) A prime number
15. (6) Divisible by 11
17. (6) Divisible by 9
18. (3) A square number
19. (3) A square number
21. (7) Its digits sum to 41
22. (3) A prime number
23. (3) Larger than 32-down
27. (5) Half of 6-down
28. (5) Divisible by 125
30. (4) Its digits are descending
31. (3) Its digits sum to 6
32. (3) A multiple of 31-down
33. (3) A multiple of 31-down
35. (3) An even number
36. (3) A square number